

Professor, Korea University – Department of Electrical and Computer Engineering, Seoul, Korea

Visiting Professor, Seoul National University Hospital, Seoul, Korea

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Educational Backgrounds

- **University of Southern California (USC) – Viterbi School of Engineering** *Los Angeles, California, USA*
 - Ph.D. (08/2009–08/2014) in **Computer Science**, Thomas Lord Department of Computer Science
Advisor: Prof. Andreas F. Molisch (Communications, Information, Learning & Quantum (CLIQ) Group (Electrical Engineering))
 - M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
 - M.S. (05/2012) in **Electrical Engineering**
 - **Korea University – College of Informatics** *Seoul, Republic of Korea*
 - M.S. (03/2004–02/2006) in **Computer Science and Engineering**
 - B.S. (03/1999–02/2004) in **Computer Science and Engineering**
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Professional Affiliations

Full-Time

- **Korea University**, *Seoul, Republic of Korea*
 - Professor (09/2019–Present)**, Department of Electrical and Computer Engineering
 - Adjunct Professor (03/2023–Present)**, Department of Communications Engineering (Samsung Electronics)
 - Adjunct Professor (03/2021–Present)**, Department of Semiconductor Engineering (SK Hynix)
 - Adjunct Professor (11/2022–Present)**, Department of Future Science and Technology Business (Graduate School)
 - Affiliated Professor (03/2025–Present)**, Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, *Seoul, Republic of Korea*
 - Assistant Professor (03/2016–08/2019)**, School of Computer Science and Engineering
- **Intel Corporation**, *Santa Clara (Silicon Valley), California, USA*
 - Systems Engineer (09/2013–02/2016)**, Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, *Los Angeles, California, USA*
 - Ph.D. Student in Computer Science (08/2009–08/2014)**, Thomas Lord Department of Computer Science
 - Annenberg Graduate Fellow (2009)**, 4-Year Full Scholarship for the Ph.D. Program in Computer Science
 - Research Assistant (01/2011–08/2014)**, Communications, Information, Learning & Quantum (CLIQ) Group
 - Teaching Assistant (01/2012–05/2013)**, Computer Science and Electrical Engineering Major Courses
- **InterDigital**, *San Diego, California, USA*
 - Intern (05/2012–08/2012)**, Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, *Seoul, Republic of Korea*
 - Research Engineer (01/2006–08/2009)**, Multimedia Research Lab, Seocho R&D Campus

Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, *Seoul, Republic of Korea*
 - Visiting Professor (10/2025–Present)**, Healthcare AI Research Institute

Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, **Advisory Committee (12/2025–Present)**
 - **National Research Foundation of Korea**, **Review Board (11/2024–10/2026)**
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University R&D/Administrative Positions

R&D Leadership Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Director (07/2024–12/2031)**, Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
 - Principal Investigator, Software Star-Lab (07/2024–12/2031)**, Quantum AI Empowered Second-Life Platform Technology

Administrative Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Associate Dean (08/2026)**, University College
 - Vice Department Chair (01/2025–08/2025)**, Academic Affairs, School of Electrical Engineering
 - Deputy Vice President (02/2022–08/2024)**, Office of Academic Affairs
 - Dean (06/2021–08/2023)**, Center for Teaching and Learning (CTL)

Committee Members

- **Korea University**, *Seoul, Republic of Korea*
 - Member (12/2023–Present)**, Council of the Interdisciplinary Program in Quantum Information Science
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Awards and Honors

Research and Academic Excellence (International)

- **Top Cited Article in ETRI Journal (Wiley), among work published in 2024** 2025
- **Finalist, AAAI Student Abstract and Poster Session – Oral Presentation** 2026 (Top 35), 2023 (Top 25)
- **IEEE ICTC Best Paper Award – IEEE Communications Society** 2022
- **Spotlight, Oral Presentation – ICML Workshop on Dynamic Neural Networks (2022)** 2022
- **IEEE MMTC Best Journal Paper Award – IEEE Communications Society** 2021
- **IEEE ICOIN Best Paper Award – IEEE Computer Society** 2021
- **IEEE MMTC Outstanding Young Researcher Award – IEEE Communications Society** 2020
- **IEEE Systems Journal Best Paper Award – IEEE Systems Council** 2020
- **Next Generation and Standards (NGS) Division Recognition Award – Intel Corporation** Q1/2015
- **Annenberg Graduate Fellowship Award – University of Southern California** 2009–2013
Awarded with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and \$120,000 Stipend (\$30,000/year for 4 years))
- **IEEE Seoul Section Student Paper Contest** Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- **IEEE VTS Seoul Chapter Award – IEEE Vehicular Technology Society** 2023, 2022, 2022, 2021, 2021, 2019

Research and Academic Excellence (Korea Regional, since 2016)

- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2025
- **Best Paper Award, The Journal of KICS – KICS** 2024
- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2023
- **Haedong Young Scholar Award – KICS and Haedong Foundation** 2018
For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

Korea University

- **Granite Tower Best Research Award (Top 3%), Totally, 2 times**
 - 2025 (Top 3%, top 53 professors at Korea University in 2025) 05/2026
 - 2024 (Top 3%, top 54 professors at Korea University in 2024) 05/2025
- **Best Research Achievement Award – Korea University, School of Electrical Engineering** 2024
- **Insung Research Grant Award – Korea University** 01/2023
For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- **Granite Tower Best Teaching Award (Top 5%), Totally, 7 times**
 - Probability and Random Process KECE209-02 (2025)
 - Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
 - Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
 - Introduction to Computers SEMI103 (2021)
 - Future Mobility Technology GEQR075 (2022)
- **Best Teaching Award (Top 20%), Totally, 7 times**
 - Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
 - Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
 - Computer Language and Laboratory EGRN151-09 (2020)
 - Object-Oriented Programming SEMI104 (2021)

Academic and University Services (International)

- **Certificate of Appreciation – IEEE Future Networks, AI/ML Working Group** 08/2025
- **Certificate of Appreciation – IEEE/IFIP WiOpt (2024)** 10/2024
- **Best Editor Award – ICT Express (Elsevier)** 07/2023
- **2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter, Awarded as a Treasure** 2022
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021
- **Certificate of Appreciation – Department of Computer Science, University of Southern California** 2010

Academic and University Services (Korea Regional)

- **Outstanding Contribution Award**
 - The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
 - Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
 - Open Standards and ICT Association (OSIA) 2025, 2021
 - The Institute of Electronics and Information Engineers (IEIE) 2025
- **Commemorative Plaque, KICS Excellent Lecture (Title: Quantum Machine Learning and Software Technologies)** 2025
- **Best Research Group Award, KICS Research Group in Quantum Communications and Quantum Computing (Group Leader)** 2025
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021

Student Awards (International)

- **Mitacs Globalink Research Award - for research in Canada, Awarded to Seok Bin Son** 2026
Project Title) Quantum Federated Learning in Distributed Systems: Challenges and Future Directions
Advisor) Prof. Trung Q. Duong (Memorial University of Newfoundland)

• Mitacs Globalink Research Award - for research in Canada, Awarded to <i>Seungcheol Oh</i>	2026
Project Title) Over the Air Quantum Federated Learning Advisor) Prof. Octavia Dobre (Memorial University of Newfoundland)	
• IEEE TCSC Award for Excellence in Scalable Computing (IEEE Computer Society), Awarded to <i>Soohyun Park</i>	2025
• Young Scientist Award (ICSANE 2025), Awarded to <i>Minji Lee</i>	2025
• IEEE Vehicular Technology Society Student Scholarship Award (2025), Awarded to <i>Gyu Seon Kim</i>	2025
• IEEE ICDCS Best Runner-Up Poster Paper Award (2025), Awarded to <i>Seok Bin Son, Soohyun Park</i>	2025

R&D Projects

University-Wide/Center/Flagship Projects

• (Center Director, PI) Net-Zero CAFE Research Center (07/2024–12/2031)	ITRC (Korea Univ)
• (SW Star-Lab, PI) Quantum AI Empowered Second-Life Platform Technology (07/2024–12/2031)	Star-Lab (Korea Univ)
• Intelligent 6G Wireless Access System Research Center (04/2021–12/2025)	6G AI Research Center (Korea Univ)
• K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks (06/2021–05/2024)	Basic Research Lab (Ajou)
• Nano UAV Intelligence Systems Research Lab (10/2020–08/2023)	ADD Military Special Research Center (Kwangwoon)
• 5G/Unmanned Vehicle Research Center (5G/UV-RC) (06/2020–12/2022)	ITRC (Hanyang)
• Human Resource Development for the Biomedical Unstructured Big Data Analysis (08/2018–12/2021)	ITRC (SNU-Hospital)
• Novel Data Science Driven Framework for Efficient Network Design (06/2017–05/2020)	Basic Research Lab (Chung-Ang)
• Intelligent Internet of Energy (IoE) Data Research Center (02/2020–05/2020)	ITRC (Kookmin)

Consortium Projects

• 6GARROW: 6G AI-Native Integrated RAN-Core Networks (09/2024–08/2027)	IITP
• AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development (04/2022–12/2026)	IITP
• Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment (04/2021–12/2025)	IITP
• Integrated Perception Technology Developments for Public Safety Platforms (06/2019–05/2023)	NRF
• Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm (07/2019–12/2021)	MHW
• Virtual Presence in Moving Objects through 5G (PriMO-5G) (06/2018–06/2021)	IITP
• Distributed Secure Platform for Scalable Clinical OMOP CDM Models (04/2019–12/2020)	MHW

Individual Projects

• Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System (03/2025–02/2028)	NRF
• Research on Trainability and Learnability in Quantum Machine Learning Models (03/2026–10/2027)	ETRI-Affiliated Institute
• SEDNA: Satellite-aided Entanglement Distribution for Networked Applications (10/2026–09/2027)	NRF
• Advancement Technology Development for Torpedo Deception Strategies in Submarines (11/2022–11/2026)	LIG Nex1
• Advancement Tech Dev for Submarine Target Identification and Engagement Support Intelligence (11/2022–11/2026)	LIG Nex1
• Research Planning for RL-based Scheduling in Autonomous Satellite Task Allocation (05/2026–10/2026)	KOITA
• mmWave-based Liquid Classification in Next-Gen Robotic Vacuum Cleaners (06/2026–09/2026)	LG Electronics (Robotics Lab)
• Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers (09/2020–11/2025)	Samsung Electronics
• LEO Satellite Routing Research using Large Language Model and Reinforcement Learning (05/2025–11/2025)	ETRI
• Discovering Quantum-Advantage Cases for Industry Adoption (09/2025–11/2025)	KTL
• Research on Generative Quantum Machine Learning Models (04/2025–10/2025)	ETRI-Affiliated Institute
• Quantum Hyper-Driving: Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility (03/2022–02/2025)	NRF
• Research on Learning-based Swarm Mission Planning Algorithms (03/2024–02/2025)	LIG Nex1
• Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks (05/2024–11/2024)	ETRI
• Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies (10/2023–09/2024)	NRF
• Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration (12/2022–04/2024)	Hyundai Motors
• NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks (09/2023–11/2023)	ETRI
• Routing Algorithms for LEO Satellite Networks (12/2022–08/2023)	Solvit System
• Optimal Positioning Algorithms for Wide-Area Relaying Networks (12/2022–08/2023)	Solvit System
• Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data (11/2022–02/2023)	Cipherome
• Cellular/Wi-Fi Handover Technology Development (02/2022–12/2022)	LG Electronics CTO Division (Smart Mobility Lab)
• Autonomous Intelligent COA Search Methods for Cyber-Attacks (12/2021–11/2022)	ADD
• Development of Quantum Deep Reinforcement Learning Algorithm using QAOA (10/2019–04/2022)	NRF
• Research Trends in Digital Twin Applications to Autonomous Driving (03/2022–04/2022)	Hyundai NGV
• Distributed Learning System Design and Implementation for Clinical Applications (02/2022–03/2022)	Cipherome
• mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving (06/2019–02/2022)	NRF
• Research on Intelligent Agent-based CPS Security and Reliability (04/2021–11/2021)	TTA
• Multi-GPU based Automotive HPC Platform Development (04/2020–10/2020)	ETRI
• Super-Resolution Performance Optimization in Mobile Platforms (05/2020–08/2020)	Samsung SDS
• Cooperative Deep Reinforcement Learning for Online Game Multi-Agents (04/2020–08/2020)	ETRI
• Deep Learning Algorithms for mVOC Concentration Analysis (03/2020–06/2020)	Samsung Electronics (C-Lab)
• Visual Recognition Software Implementation using Deep Learning Tools (05/2019–11/2019)	Hyundai Motors
• Verification Testbed Implementation for Privacy-Preserving Trust Data Generation (10/2019–11/2019)	ETRI
• Measurement and Analysis of Multi-Task GPU Scheduling Delays (05/2019–10/2019)	ETRI

- *mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services* (06/2016–05/2019) **NRF**
- *Probabilistic Decision Making and Econometric Methods for Micro-Grid* (05/2017–04/2019) **KEPCO Research Institute**
- *GPU Scheduling Performance Analysis under Queueing Delay Considerations* (05/2018–10/2018) **ETRI**
- *Improving Massive Deep Learning Training via Computation and Communication Acceleration* (04/2018–10/2018) **ETRI**
- *A Priori Techniques Research for Efficient Multi-Edge Computing* (06/2017–12/2017) **Samsung Electronics (Software Center)**
- *Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms* (04/2017–12/2017) **IITP**
- *Parsing Techniques for Artificial Neural Network (ANN) Data Processing* (09/2017–11/2017) **ETRI**

University of Southern California, Selected Projects

- *Video Aware Wireless Networks (VAWN) Research Program* **Intel Labs, Verizon Wireless, Cisco Systems**
- *60 GHz Real-Time Wireless Video Broadcasting* **Disney Research Zürich**
- *Annenberg Graduate Fellowship Award* **University of Southern California**

Korea University (KU), Selected Projects

- *Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control* (08/2025–07/2026) **PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking Univ)**
- *AI Teaching Assistant Research for LLM-based Large-Scale Education* (10/2024–02/2025) **KU University College**
- *Autonomous Mobility Control using Quantum Deep Learning* (03/2023–02/2024) **KU Insung Research Grant**
- *Mobile Access Algorithm Design using Economic Theory and AI* (03/2020–02/2021) **KU Future Research Grant**

Selected Publications

- **12,417+ Citations** (H-index: 54+, i10-index: 257+), obtained from Google Scholar Profile (as of September 25, 2026)
- **Journals and Magazines: 180** publications; among them, **137** IEEE/Nature/APS publications

Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) (Editors: X. Lin, J. Zhang, Y. Liu, [J. Kim](#))

Selected Papers

■ Top-Tier Conference Publications

- [ACCV'26] Personalized Quantum Federated Visual Learning via Raced Architecture Search and Ensemble Calibration
ACCV 2026 (Osaka, Japan, December 2026) (S. B. Son, S. Park, [J. Kim](#), W. Lee)
- [CIKM'26] Strong Lottery Ticket Hypothesis for Quantum Neural Networks
ACM CIKM 2026 (Rome, Italy, November 2026) (H. Kwon, H. K. Lee, [J. Kim](#), S. W. Yoon)
- [CIKM'26] Beneath the Noise: Uncovering and Distilling the Encoding Bottleneck of Quantum Models
ACM CIKM 2026 (Rome, Italy, November 2026) (S. Lee, C. Kim, S. Son, [J. Kim](#), H. In)
- [ICML'26] Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Speed-Up in Query Complexity
ICML 2026 (Seoul, Korea, July 2026) (H. K. Lee, [J. Kim](#), S. W. Yoon)
- [ACL'26] Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom
ACL 2026 (San Diego, CA, USA, July 2026) (J. Lee, G. Youn, S. Lee, C. Im, [J. Kim](#), C. Yoo)
- [CIKM'25] Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models
ACM CIKM 2025 (Seoul, Korea, November 2025) (E. J. Roh, [J. Kim](#))
- [CIKM'25] Filtered One-Shot Training for Quantum Architecture Search
ACM CIKM 2025 (Seoul, Korea, November 2025) (S. B. Son, S. Y.-C. Chen, [J. Kim](#), S. Park)
- [CIKM'25] LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data
ACM CIKM 2025 (Seoul, Korea, November 2025) (G. Youn, J. Lee, [J. Kim](#), C. Yoo)
- [IPDPS'25] AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming
IEEE IPDPS 2025 (Milan, Italy, June 2025) (X. Piao, J. Shim, [J. Kim](#), J. Kim)
- [WiOpt'25] Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach
IEEE/IFIP WiOpt 2025 (Linköping, Sweden, May 2025) (G. S. Kim, J. Chung, T. Q. Duong, S. Park, [J. Kim](#))
- [CIKM'24] Hands-On Introduction to Quantum Machine Learning
ACM CIKM 2024, Tutorial (Boise, ID, USA, October 2024) (S. Y.-C. Chen, [J. Kim](#))
- [WiOpt'24] Advanced Taxing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management
IEEE/IFIP WiOpt 2024 (Seoul, Korea, October 2024) (S. Lee, G. S. Kim, S. Park, [J. Kim](#))
- [CIKM'23] Quantum Split Learning for Privacy-Preserving Information Management
ACM CIKM 2023 (Birmingham, United Kingdom, October 2023) (S. Park, H. Baek, [J. Kim](#))
- [CIKM'23] Logarithmic Dimension Reduction for Quantum Neural Networks
ACM CIKM 2023 (Birmingham, United Kingdom, October 2023) (H. Baek, S. Park, [J. Kim](#))
- [AAAI'23] Quantum Multi-Agent Meta Reinforcement Learning
AAAI 2023 (Washington DC, USA, February 2023) (W. Yun, J. Park, [J. Kim](#))
- [CIKM'22] Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly
ACM CIKM 2022 (Atlanta, GA, USA, October 2022) (W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, [J. Kim](#))

- [WiOpt'22] Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution *IEEE/IFIP WiOpt 2022* (Torino, Italy, September 2022) (M. Choi, W. Yun, J. Kim)
- [INFOCOM'22] Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks *IEEE INFOCOM 2022* (Virtual, May 2022) (H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim)
- [ICDCS'20] Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications *IEEE ICDCS 2020* (Virtual, November/December 2020) (Ü. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen)
- [IJCAI'19] Randomized Adversarial Imitation Learning for Autonomous Driving *IJCAI 2019* (Macau, China, August 2019) (M. Shin, J. Kim)
- [ICDCS'18] ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture *IEEE ICDCS 2018* (Vienna, Austria, July 2018) (S. Ahn, J. Kim, E. Lim, W. Choi, A. Mohaisen, S. Kang)
- [ACM-MM'17] REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints *ACM Multimedia 2017* (Mountain View, CA, USA, October 2017) (J. Koo, J. Yi, J. Kim, M. Hoque, S. Choi)
- [MobiSys'10] Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones *ACM MobiSys 2010* (San Francisco, CA, USA, June 2010) (J. Paek, J. Kim, R. Govindan)

■ Magazines and Journals

◀ Accepted ▶

- [CIM.accepted] Rethinking Privacy-Preserving Distributed Learning in Quantum Hilbert Spaces, *IEEE Computational Intelligence Magazine*, v(n):ppp-ppp (S. B. Son, J. Kim, S. Park)
- [CM.accepted] Living on the Quantum Edge: Agentic Quantum AI for Mobile Networks, *IEEE Communications Magazine*, v(n):ppp-ppp (H. Ahn, H. Lee, S. Jung, M. Bennis, S. Park, J. Kim)
- [CM.accepted] Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective, *IEEE Communications Magazine*, v(n):ppp-ppp (G. S. Kim, J. Chung, S. Park, S. Jung, T. Q. Duong, J. Kim)
- [NM.accepted] Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks, *IEEE Network*, v(n):ppp-ppp (H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, J. Kim)

◀ 2026 ▶

- [AC'26.12] Physics-Validated Action Masking with LLM-Guided Decomposition for Autonomous Excavation, *Automation in Construction*, 192:107209 (J. Cho, M. Shin, B. Kim, B. Cha, J. Kim, S. Jung)
- [TMC'26.11] Scale-Reconfigurable Multi-Agent Reinforcement Learning for Aerial Networks in Dynamic Environments, *IEEE Transactions on Mobile Computing*, 25(11):ppp-ppp (G. S. Kim, E. J. Roh, S. Jung, S. Park, J. Kim)
- [TMC'26.11] Efficient Quantum Neural Architecture Search with Filtered Supernet for Mobile Learning Systems, *IEEE Transactions on Mobile Computing*, 25(11):ppp-ppp (S. B. Son, H. Ahn, S. Y.-C. Chen, S. Park, J. Kim)
- [IOTJ'26.11] Verifiable Quantum Machine Learning Inference Consistency for Classical Clients in Noisy Quantum Clouds, *IEEE Internet of Things Journal*, 13(22):ppp-ppp (S. Lee, S. B. Son, C. Kim, J. Kim, H. P. In)
- [INM'26.10] Toward Quantum Combinatorial Optimization, *IEEE Nanotechnology Magazine*, 20(5):ppp-ppp (Y. Hong, J. Gil, D. Kim, S. Oh, B. S. Kim, J. Kim)
- [TON'26.mm] Differentially-Private Quantum Federated Learning Against Gradient Inversion Attacks, *IEEE Transactions on Networking*, 34:ppp-ppp (C. Im, S. Park, J. Kim)
- [IOTJ'26.10] USD2C: UAV-Assisted Secure Data Collection and Computation Framework for Internet of Multimedia Things (IoMT) Applications, *IEEE Internet of Things Journal*, 13(19):45810–45823 (I. Ahmad, J. Kim, S. Shin)
- [TVT'26.10] Time-to-Collision Aware Autonomous Driving Motion Control under Risky Scenario Considerations, *IEEE Transactions on Vehicular Technology*, 75(10):ppp-ppp (S. Kim, J. Kim, J. Kim, S. Jung)
- [IOTJ'26.09] Exponentially Scalable Quantum Reinforcement Learning for Large-Scale Network Scheduling, *IEEE Internet of Things Journal*, 13(18):43910–43925 (G. S. Kim, E. J. Roh, S. Park, J. Kim)
- [Access'26.09] Transformer-based Offline Reinforcement Learning for Intelligent Torpedo Evasion in Autonomous Underwater Vehicles, *IEEE Access*, 14:135721–135736 (H. Ahn, H. Lee, B. Kang, J. Choi, Y. Yoon, J. Kim)
- [FGCS'26.09] GraMA: A Gradient Matrix-Guided Assignment Method for Solving Qubit Mapping Problems, *Future Generation Computer Systems*, 182:108485 (X. Piao, J. Kim, J.-K. Kim)
- [CM'26.08] Sustainable 6G Non-Terrestrial Networks: Enabling Net-Zero Operation via Renewable Energy Integration and Intelligent Control, *IEEE Communications Magazine*, 64(8):110–116 (H. Lee, S. Park, S. Jung, J. Kim)
- [TON'26.08] Hallucination-Aware Hierarchical LLM for Autonomous UAV Mobility Control: A Safe Reinforcement Learning Approach, *IEEE Transactions on Networking*, 34:6827–6842 (H. Ahn, S. Oh, G. S. Kim, S. Park, S. Jung, J. Kim)
- [IOTJ'26.08] Learning-based Resilient Dynamic Routing for Fault-Tolerant Robust LEO Satellite Networks, *IEEE Internet of Things Journal*, 13(16):37402–37414 (G. S. Kim, C. Im, Y. G. Kim, J. Ha, J. Kim)
- [EAAI'26.08] A Comprehensive Review on Quantum Deep Neural Networks for Prognostics and Health Management: Fundamentals, Challenges and Opportunities, *Engineering Applications of Artificial Intelligence*, 177:114991 (M. Jha, S. Chen, C. Kulkarni, J. Kim)
- [TMC'26.07] Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient Cube-Satellite Networks, *IEEE Transactions on Mobile Computing*, 25(7):10385–10401 (S. Park, G. S. Kim, Y. Cho, S. Jung, Z. Han, J. Kim)
- [Access'26.07] Two-Stage MPC-Guided Imitation and Reinforcement Learning for Autonomous Underwater Vehicle Control in Dynamic Obstacle Environments, *IEEE Access*, 14:99636–99648 (S. Kim, E. J. Roh, I. Song, J. Kwon, S. Kwon, Y. Kim, J. Kim)
- [IOTJ'26.06] Quantum Federated Gradient Aggregation Using the Parameter-Shift Rule, *IEEE Internet of Things Journal*, 13(11):25367–25379 (J. Chung, C. Im, S. Park, J. Kim, W. Lee)
- [Nature'26.05] Fourier Analysis Perspective on Quantum Neural Networks, *Communications Physics (Nature)*, 9:176 (S. Oh, E. J. Roh, A.

- [TVT'26.04] Large-Scale Battery-Conscious Collision-Free Path-Finding for Sustainable and Autonomous Multi-AGV Mobility Control, *IEEE Transactions on Vehicular Technology*, 75(4):5364–5375 (Y. Cho, S. Ahn, S. Park, J. Kim)
- [TAES'26.03] Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket, *IEEE Transactions on Aerospace and Electronic Systems*, 62:3838–3863 (G. S. Kim, J. Chung, S. Jung, S. Park, J. Kim)
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■ IEEE ComSoc, VTS, and Computer Society Conference Contributions

- 50+ papers at INFOCOM, ICDCS, GLOBECOM, ICC, WCNC, CCNC, PIMRC, VTC, APWCS, IEEE Quantum Week
- 100+ papers at other *IEEE ComSoc*, *IEEE VTS*, *IEEE Computer Society*, and *ACM* conferences

Academic Activities (Selected: Quantum AI and Quantum Computing)

Invited Talks at Universities (Selected)

- Quantum Neural Networks: Algorithms and Applications
[Keio University](#) (Tokyo, Japan, 12/2025)
- Quantum Reinforcement Learning: From Foundations to Emerging Applications
[Ulsan National Institute of Science and Technology \(UNIST\)](#) (Ulsan, Korea, 12/2025)
- Quantum Reinforcement Learning: Algorithms and Applications
[Indiana University, Bloomington](#) (Bloomington, Indiana, USA, 10/2025)
- Quantum Neural Networks: Theory, SW, and Applications
[Gwangju Institute of Science and Technology \(GIST\)](#) (Gwangju, Korea, 05/2025)
- Quantum Neural Networks: Theory, SW, and Applications
[Seoul National University Hospital](#) (Seoul, Korea, 03/2025)
- Quantum Neural Networks: Theory, SW, and Applications
[Sungkyunkwan University](#) (Suwon, Korea, 01/2025)
- Quantum Neural Networks: Theory, SW, and Applications
[Yonsei University](#) (Seoul, Korea, 09/2023)

Journal Special Issues (Guest Editorials)

- Quantum Technologies for Communication Systems (2025), *IEEE/KICS Journal of Communications and Networks*
- Advances in Quantum Networks: Enabling Computing, Security, and AI Innovations (2025), *Discover Networks (Springer)*
- Quantum Algorithms, Optimization, and A.I. (2025), *The Journal of Supercomputing (Springer)*

Organizing Committee Contributions for International Conferences, Symposiums, and Workshops

- Quantum Neural and Reinforcement Learning for Adaptive Sequential Intelligence (ID 47)
[IEEE QAI 2026](#) (Nottingham, United Kingdom, 12/2026)
- Workshop on Quantum Computing and Reinforcement Learning (2025)
[IEEE Quantum Week \(QCE\) 2026](#) (Toronto, Ontario, Canada, 09/2026)
- Quantum Technologies for Earth Observation (2026)
[IEEE IGARSS 2026](#) (Washington DC, USA, 08/2026)
- Special Session: Quantum Machine Learning Algorithms and Applications (2026)
[IEEE WCCI 2026](#) (Maastricht, The Netherlands, 06/2026)
- Workshop on Quantum Machine Learning in Signal Processing and Artificial Intelligence (2026)
[IEEE ICASSP 2026](#) (Barcelona, Spain, 04/2026)
- Workshop: Quantum Computing - Artificial Intelligence (QC-AI) for Industry Applications and Scientific Discovery (2026)
[IEEE QCNC 2026](#) (Kobe, Japan, 04/2026)
- Quantum Computing for Classical Computational Challenges in HPC (2026)
[IEEE QCNC 2026](#) (Kobe, Japan, 04/2026)
- Workshop on Quantum Machine Learning (2026)
[IEEE ICOIN 2026](#) (Hanoi, Vietnam, 01/2026)
- Workshop on Quantum Computing for Communications and Learning (2025)
[IEEE GLOBECOM 2025](#) (Taipei, Taiwan, 12/2025)
- Workshop on Quantum Computing and Reinforcement Learning (2025)
[IEEE Quantum Week \(QCE\) 2025](#) (Albuquerque, New Mexico, USA, 09/2025)
- Workshop on Quantum Machine Learning in Signal Processing and Artificial Intelligence (2025)
[IEEE ICASSP 2025](#) (Hyderabad, India, 04/2025)
- Workshop on Quantum Computing and Reinforcement Learning (2024)
[IEEE Quantum Week \(QCE\) 2024](#) (Montreal, Quebec, Canada, 09/2024)
- Workshop on Quantum Algorithms, Optimization, and Artificial Intelligence (2024)
[IJCAI 2024](#) (Jeju, Korea, 08/2024)

Program Committee Contributions for International Conferences, Symposiums, and Workshops

- Workshop on Secure and Trustworthy Quantum Machine Learning (2026)
[NeurIPS 2026](#) (Atlanta, Georgia, USA, 12/2026)
- Workshop on Quantum Machine Learning (2026)
[IEEE ICOIN 2026](#) (Hanoi, Vietnam, 01/2026)

- Workshop on Quantum Computing for Communications and Learning (2025)
IEEE GLOBECOM 2025 (Taipei, Taiwan, 12/2025)
- Workshop on Quantum Data and Machine Learning: Systems, Theory and Hardware (2025)
IEEE ICDE 2025 (Hong Kong, China, 05/2025)
- Workshop on Quantum Computing and Artificial Intelligence (2024)
AAAI 2024 (Philadelphia, Pennsylvania, USA, 02/2024)

International Tutorials and Speeches

- Hands-On Quantum AI for Sequential Decision-Making: Variational Quantum Models, Quantum Reinforcement Learning, and Self-Programming Quantum Circuits (ID 46)
IEEE QAI 2026 Tutorial (Nottingham, United Kingdom, 12/2026)
- Introduction to Distributed Quantum Artificial Intelligence: Neural Networks, Reinforcement Learning, and Multi-Agent Systems
IEEE WCCI 2026 Tutorial (Maastricht, The Netherlands, 06/2026)
- Quantum-Enhanced Learning for Future Communication Networks
IEEE GLOBECOM 2025 Tutorial (Taipei, Taiwan, 12/2025)
- Quantum Reinforcement Learning: From Foundations to Emerging Applications
IEEE QAI 2025 Tutorial (Naples, Italy, 11/2025)
- Quantum Machine Learning: Algorithms and ICT Convergence Applications
IEEE ICTC 2025 Invited Talk (Jeju, Korea, 10/2025)
- Hands-On Introduction to Quantum Machine Learning
IEEE QCE 2025 Tutorial (Albuquerque, New Mexico, USA, 09/2025)
- Quantum Reinforcement Learning for Computer Networks and Autonomous Mobility: Modern Trends and Future Directions
IEEE VTS APWCS 2025 Invited Speech (Tokyo, Japan, 08/2025)
- Quantum Reinforcement Learning: Algorithms and Applications
IEEE Future Networks Webinar (Future Networks AI/ML and QIT Working Groups) (Webinar, 08/2025)
- Introduction to Quantum Machine Intelligence
IEEE SSCI 2025 Tutorial (Trondheim, Norway, 03/2025)
- Hands-On Introduction to Quantum Machine Learning
ACM CIKM 2024 Tutorial (Boise, Idaho, USA, 10/2024)
- Quantum Reinforcement Learning: Concepts, Models, and Applications
ICIT 2024 Keynote Speech (Danang, Vietnam, 09/2024)
- A Paradigm Shift in Future Networks with Quantum Deep Learning
IEEE ICUFN 2022 Tutorial (Barcelona, Spain, 07/2022)

Academic Activities and Research Supervision

Research Supervision

■ Ph.D. Alumni

1. Soohyun Park (03/2019–08/2023 (PhD), 09/2023–02/2024 (Postdoc)), **Sookmyung Women's University (Professor)**
2. Hankyul Baek (03/2021–02/2024 (PhD), 03/2024–03/2025 (Postdoc)), ETRI AI Safety Institute (Researcher)
3. Hyunsoo Lee (03/2021–02/2026 (PhD)), LG Electronics (Researcher)
4. Gyu Seon Kim (03/2023–02/2026 (PhD), 03/2026–04/2026 (Postdoc)), RealtimeVisual (Researcher, *Military Service Exception*)
5. Seok Bin Son (03/2022–02/2027 (PhD), 03/2027–Present (Postdoc)), Korea University (Postdoctoral Scholar)
6. Emily Jimin Roh (03/2024–02/2027 (PhD), 03/2027–Present (Postdoc)), Korea University (Postdoctoral Scholar)
7. Yunpyo Hong (03/2026–02/2028 (Expected) (PhD))

■ M.S. Alumni

1. Kyeongseon Kim (09/2017–08/2019), KAIST (Ph.D. Student in Computer Science)
2. Dohyun Kwon (03/2018–02/2020), Hyundai Marine & Fire Insurance (Researcher)
3. Dohyun Kim (03/2018–02/2020), Naver Webtoon (Researcher)
4. MyungJae Shin (03/2018–02/2020), Naver (Researcher)
5. Jaeho Choi (03/2019–02/2021), Korea Meteorological Administration (Researcher, *Military Service Exception*)
6. Yoo Jeong (Anna) Ha (03/2021–02/2023), The University of Chicago (Ph.D. Student in Computer Science)
7. Jaehyun Chung (09/2023–08/2025), Hanwha Aerospace (Researcher)
8. Chaemoon Im (03/2024–02/2026), Korea University Net-Zero CAFE Research Center (Researcher)
9. Yeryeong Cho (03/2024–02/2026), Korea University (Ph.D. Student in Electrical and Computer Engineering)
10. Hyojun Ahn (03/2025–02/2027) Korea University (Ph.D. Student in Electrical and Computer Engineering)

■ Former Research Professors and Postdoctoral Scholars

- Minseok Choi (09/2018–02/2019, jointly with Prof. Andreas F. Molisch (USC)), **Kyung Hee University (Professor)**
- Soyi Jung (03/2021–08/2021, jointly with Prof. Marco Levorato (UC-Irvine)), **Ajou University (Professor)**
- Ju-Hyung Lee (03/2022–02/2023, jointly with Prof. Andreas F. Molisch (USC)), Nokia USA (Researcher)

■ Former Full-Time Researchers

- Rhoan Lee, LG CNS (Researcher) (M.S. in Statistics, Columbia University in the City of New York)
- Daeyeun Kim (07/2025–07/2026), Korea University (Ph.D. Student in Electrical and Computer Engineering)
- Donggun Lee (04/2026–08/2026), The University of Michigan (B.S. Student in Data Science)

IEEE Activities (Membership, Editorial Boards, and Services)

- Senior Member (2018–), Member (2006–2017)
- Editor (2026–), **IEEE Transactions on Machine Learning in Communications and Networking** (Area 1: Learning for PHY/MAC Communications & Signal Processing)

- *Editor* (2026–), **Quantum Machine Intelligence** (Springer-Nature)
 - *Associate Editor* (2025–), **ACM Computing Surveys**
 - *Associate Editor* (2025–), **IEEE Communications Surveys and Tutorials**
 - *Editor* (2023–), **IEEE Internet of Things Journal**
 - *Associate Editor* (2020–), **IEEE Transactions on Vehicular Technology** (Area: Vehicular Electronics and Systems)
 - *Guest Editor*, **Journal of Communications and Networks** (S.I. on Quantum Technologies for Communication Systems)
 - *Guest Editor*, **IEEE Communications Standards Magazine** (S.I. on Recent and Future Evolution of Wi-Fi)
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References

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